

Mechanical Motion-Capture Glove for Robotics (Variants)

The group of inventions relates to the field of robotics and mechatronics, specifically to wearable exoskeletal devices. The objective of the present invention is to develop a mechanical glove that is kinematically identical to the anatomical structure of the human hand, designed for simultaneous real-time control of a robotic hand and for collecting kinematic parameters of the user's finger movements for the purpose of subsequent training and programming of robotic systems.

Modern control systems for robotic manipulator-type hands can be implemented in various ways, including the use of pre-developed software interfaces, stationary control panels or joysticks, as well as visual tracking systems that monitor finger movements. However, existing methods are often limited in accuracy and incapable of reproducing complex and fine finger motions, or they tend to be bulky and require specific operating conditions. These limitations significantly reduce their applicability for precise manipulation of complex objects. Therefore, there is a need for a device that simultaneously provides full kinematic correspondence to human fingers, high sensitivity and accuracy, as well as comfort and mobility during use. A mechanical glove of this kind enables the operator to intuitively control a robotic hand in real time and also allows for the collection of detailed kinematic data and force feedback information for training and programming robotic systems.

An invention is known from patent WO 2020088015A1 titled "Hand Motion Capture Device." The essence of the device lies in a hand motion capture apparatus that includes a base and a plurality of linkage structures. The device further comprises: A control unit positioned on the base, the control unit being configured to transmit the obtained detection signal to a host computer and receive one or more control signals transmitted back from the host computer; A first drive unit with tactile (force) feedback, connected between the base and the linkage structure. The first drive unit is configured to detect the rotation angle of the linkage structure relative to the base, generate a detection signal, output the detection signal to the control unit, and, upon receiving a corresponding feedback control signal from the control unit, output tactile (force) feedback in accordance with said signal; A second drive unit with tactile (force) feedback, located at a joint of the linkage structure. The second drive unit is configured to detect

the rotation angle of the corresponding link, generate a detection signal, output the detection signal to the control unit, and, upon receiving a corresponding feedback control signal from the control unit, output tactile (force) feedback in accordance with said control signal.

The drawbacks of the known device include the following:

- The use of protruding mechanical components above the fingers increases the risk of damage during active use and may interfere with interaction with external objects;
- The absence of tactile sensors at the fingertips significantly limits the device's capability to collect tactile information;
- Incomplete detection of all degrees of freedom of the finger reduces gesture recognition accuracy and restricts the device's field of application.

An invention is known from patent CN 111941401A titled "Highly Universal Exoskeleton Glove for Motion Capture." The invention relates to a universal exoskeleton glove for motion capture, characterized by the inclusion of: a glove, a wrist bracelet, a controller, a control panel, a connecting mechanism, a pressure sensor, a telescopic displacement sensor, and a motor. The glove comprises: a palm portion, a finger portion, and a joint portion. The finger portion includes: one two-segment finger and four three-segment fingers. The two-segment finger consists of: a fingertip segment one and a base segment one. The three-segment fingers consist of: a fingertip segment two, a base segment two, and a middle segment. Two adjacent joints are rotatably connected via the joint portion, which includes a first fixing block and a second fixing block, with each block fixed to adjacent phalanges of the finger. These blocks are rotatably connected via a motor. The finger segments are interconnected using the connecting mechanism. During relative movement of the palm, fingers, and joints, the connecting mechanism collects motion data of the hand and transmits it to the motor for feedback.

The drawbacks of the known device include the following:

- The use of protruding mechanical components above the fingers increases the risk of mechanical failure during intensive use and may obstruct interaction with external objects;
- The absence of tactile sensors at the fingertips significantly limits the device's capability to collect tactile information;
- Incomplete detection of all degrees of freedom of the fingers reduces the accuracy of gesture recognition and limits the applicability of the device.

An invention is known from patent US 20180311570A1 titled "Motion Capture Glove Device with Exotendons and Compression-Based Haptic Feedback." The essence of the invention lies in a glove system for motion capture and tactile feedback, comprising: a. A glove portion worn on the dorsal side of the user's hand, the glove having individual sections for the

fingers and thumb; b. A plurality of securing caps covering the fingertips; c. Multiple attachment points designed to generate sensor data that determine finger flexion/extension and abduction/adduction; d. A plurality of tendon-like cables for transmitting data on finger flexion/extension and abduction/adduction to a set of measuring devices for subsequent processing, wherein the cables are flexible and allow for mobility; e. Several motors that maintain constant tension in the cables and enable retraction of the fingers and thumb in response to virtual stimuli; f. A housing structure located on the forearm and connected to the glove via said cables, the housing comprising at least one motor unit and one routing (guiding) system.

The drawbacks of the known device include the following:

- The device functions correctly only when the wrist remains stationary. During wrist flexion, the system may misinterpret finger positions, resulting in distorted gesture data or inaccurate grasp representation;
- The absence of tactile sensors at the fingertips significantly limits the device's ability to collect tactile information;
- Lateral finger movements (e.g., separation of the index and middle fingers) are not captured, which prevents realistic modeling of grasping actions and recognition of natural gestures.

The identified analogues coincide with the claimed technical solution only in certain individual features; therefore, no single prototype has been identified, and the claims have been drafted without a preamble (restrictive part).

The technical problem addressed by the present invention, and the corresponding technical result achieved, is the development of a Mechanical Motion-Capture Glove for Robotics (Variants), which provides for the implementation of:

- Detection of wrist flexion, which is essential for accurately calculating finger joint angles while accounting for changes in palm orientation in space;
- Collection of tactile information from the fingertips, enabling the measurement of pressure force, touch, and interaction with objects;
- Full motion capture of the fingers, including abduction and adduction, with the ability to reconstruct individual trajectories of each finger.

The essence of the claimed technical solution is a Mechanical Motion-Capture Glove for Robotics, designed to be worn on the human wrist, **comprising:** for each finger A first fastener connecting the distal and intermediate phalanges of the finger, wherein a tactile sensor is installed in the area of the distal phalanx of the first fastener, configured to measure the pressure force applied by the fingertip during contact with objects; a cable attachment is provided on the

dorsal side of the phalanx of the first fastener, enabling the transmission and measurement of flexion/extension of the distal and intermediate phalanges; A second fastener mounted on the proximal phalanx of the finger, with a cable attachment on the dorsal side of the proximal phalanx, enabling the transmission and measurement of flexion/extension of the proximal phalanx; A rigid mechanical coupling is provided on the second fastener to transmit mechanical motion to a first encoder, which is configured to measure the rotation angle of the proximal phalanx relative to the base of the hand, thus enabling feedback during grasping and manipulation; For each finger, a first encoder is located on the dorsal side of the hand and is intended to measure finger abduction and adduction; The measurement is carried out via a rigid mechanical connection between the first encoder and the second fastener mounted on the proximal phalanx, enabling the transmission of angular displacement during lateral finger movements and precise tracking of the finger's position in the plane perpendicular to flexion/extension; For the thumb, an additional second encoder is provided, mounted on the dorsal side of the hand, configured to measure the rotation of the thumb relative to the base of the hand, taking into account its anatomical mobility — including rotation, opposition, and abduction in the saddle joint — thereby providing an additional degree of freedom in the measurement system and enabling accurate reproduction of thumb movements during object grasping and manipulation; At the transition between the hand and wrist, a branching mechanism is located, designed to distribute the cables running from the phalanges to the actuation and measurement unit mounted on the forearm; The mechanism ensures proper routing of cables from each finger, preventing cable crossing and excessive tension during hand movements, as well as reducing friction and wear through the use of guiding channels and rotary elements; A third encoder is mounted on the wrist to measure flexion and extension at the wrist joint, enabling the recording of the angular position of the hand relative to the forearm; This allows for accounting of the wrist's position when calculating the overall position of the fingers and hand, ensuring the correction of finger flexion measurements based on wrist angle; A third fastener is positioned on the forearm, where a set of encoders is installed; The rotation of each encoder in this set is mechanically linked to finger movement via cables routed through the branching mechanism; Each encoder in the set is configured to measure: the angular displacement occurring during flexion or extension of a specific phalanx of a finger; the direction and magnitude of movement transmitted through the cable system from the finger to the respective encoder. A Mechanical Motion-Capture Glove for Robotics, configured to be worn on a human wrist, **comprising:** For each finger, a first fastener connecting the distal and intermediate phalanges of the finger, wherein a tactile sensor is installed in the region of the distal phalanx of

the first fastener, the sensor being configured to measure the pressure force applied by the fingertip during contact with objects; A cable attachment provided on the dorsal side of the phalanx of the first fastener, configured to transmit and measure the flexion and extension of the distal and intermediate phalanges; For each finger, a second fastener mounted on the proximal phalanx, wherein a cable attachment is provided on the dorsal side of the proximal phalanx of the second fastener, configured to transmit and measure the flexion and extension of the proximal phalanx; The second fastener includes a rigid connection for transmitting mechanical movement to a first encoder, the encoder being configured to measure the rotation angle of the proximal phalanx relative to the base of the hand, thereby providing feedback during grasping and manipulation; The first encoder for each finger is located on the dorsal surface of the hand and is configured to measure finger abduction and adduction, the measurement being enabled by a rigid mechanical connection between the first encoder and the second fastener mounted on the proximal phalanx, configured to transmit angular displacement during lateral movements of the finger and ensure accurate registration of its position in the plane perpendicular to flexion and extension; For the thumb, an additional second encoder is mounted on the dorsal side of the hand and is configured to measure the rotation of the thumb relative to the base of the hand, taking into account its anatomical mobility, including rotation, opposition, and abduction in the saddle joint, thereby providing an additional degree of freedom in the measurement system and enabling accurate reproduction of thumb movements during grasping and manipulation; A branching mechanism is positioned at the transition between the hand and the wrist, configured to distribute cables routed from the phalanges to the actuation and measurement unit located on the forearm, and to ensure cable routing from each finger while avoiding cable crossing and excessive tension during wrist movement, as well as reducing friction and wear by means of guiding channels and rotary elements; A third encoder is mounted on the wrist and is configured to measure flexion and extension at the wrist joint, to determine the angular position of the hand relative to the forearm, account for the influence of wrist position on the overall position of the fingers and hand, and provide correction of finger flexion calculations based on wrist flexion angle; A third fastener is located on the forearm and carries a set of encoders, each of which is mechanically linked via cables—routed through the branching mechanism—to a corresponding finger, each encoder in the set being configured to measure: the angular displacement occurring during flexion or extension of a specific phalanx; The direction and magnitude of movement transmitted through the cable system from the finger to the corresponding encoder; To enable force feedback functionality, motors are additionally installed on the encoders of the set mounted on the third fastener on the forearm, each motor being rigidly coupled to the shaft of the corresponding

encoder and thus to the cable transmitting motion to the phalanx, thereby enabling controlled resistance to finger movement for simulating interaction with virtual or physical objects, and allowing forced stopping or slowing of finger motion.

The claimed technical solution is illustrated in Fig. 1 through 3:

Fig. 1 shows a schematic view of the mechanical glove (view of the hand);

Fig. 2 shows a schematic general view of the mechanical glove with motors;

Fig. 3 shows a schematic general view of the mechanical glove without motors.

The reference numerals used in the figures denote the following components:

- 1 – first fastener;
- 2 – tactile sensor;
- 3 – second fastener;
- 4 – first encoder;
- 5 – second encoder;
- 6 – branching mechanism;
- 7 – third encoder;
- 8 – third fastener;
- 9 – set of encoders;
- 10 – motor.

The following describes a first embodiment of the mechanical motion-capture glove without force feedback.

The first embodiment of the claimed mechanical motion-capture glove for robotics is designed to be worn on the human wrist and is configured to measure and transmit kinematic parameters of finger flexion/extension and abduction/adduction movements, as well as to collect tactile information from the fingertips and to detect wrist flexion/extension. The glove comprises the following functional components.

The first fastener (1) connects the distal and intermediate phalanges of the finger, forming a rigid or semi-rigid structure. A tactile sensor (2) is installed in the area of the distal phalanx of the first fastener (1), configured to measure the pressure force applied by the fingertip during contact with objects. On the dorsal side of the phalanx of the first fastener (1), a cable attachment is provided, through which flexion and extension of the distal and intermediate phalanges are transmitted and measured.

The second fastener (3) is mounted on the proximal phalanx. On the dorsal side of the proximal phalanx of the second fastener (3), a cable attachment is provided, by means of which

the flexion and extension of the proximal phalanx of the finger are transmitted and measured. In addition, the second fastener (3) includes a rigid coupling designed to transmit mechanical motion to the first encoder (4). The first encoder (4) is configured to measure the rotation angle of the proximal phalanx relative to the base of the hand, thereby providing feedback during grasping and manipulation.

The first encoder (4) is located on the dorsal surface of the hand and is intended for measuring finger abduction and adduction (i.e., lateral spreading and convergence relative to the axis of the hand). The measurement is performed via a rigid mechanical connection between the first encoder (4) and the second fastener (3) mounted on the proximal phalanx of the finger. This structural solution ensures the transmission of angular displacement during lateral finger movements, allowing precise registration of the finger's position in the plane perpendicular to flexion and extension.

An identical structure, as described above, is provided for the remaining fingers of the hand. This enables comprehensive measurement of all primary degrees of freedom of each finger, including: flexion and extension of the distal and intermediate phalanges; flexion and extension of the proximal phalanx; abduction and adduction (movement in the lateral plane relative to the axis of the hand); and measurement of pressure applied to the fingertip using integrated tactile sensors.

An additional second encoder (5) is provided for the thumb and is mounted on the dorsal side of the hand. The second encoder (5) is configured to measure the rotation of the thumb relative to the base of the hand, enabling consideration of its unique anatomical mobility, including movement in the saddle joint—such as rotation, opposition, and abduction. The installation of the second encoder (5) provides an additional degree of freedom in the measurement system and allows for more accurate reproduction of thumb movements during grasping and manipulation of objects.

A branching mechanism (6) is located at the transition between the hand and the wrist. It serves to distribute the cables running from the finger phalanges to the actuation and measurement unit positioned on the forearm. The branching mechanism (6) ensures proper routing of the cables from each finger in such a way as to prevent cable crossing and excessive tension during wrist movement, while also reducing friction and wear through the use of guiding channels and rotary elements.

A third encoder (7) is mounted on the wrist and is configured to measure flexion and extension of the hand at the wrist joint. The third encoder (7) records the angular position of the hand relative to the forearm, allowing for the influence of wrist position on the overall

configuration of the fingers and the entire hand to be taken into account. The presence of this sensor—i.e., the third encoder (7)—enables correction of finger flexion calculations based on the wrist flexion angle.

A third fastener (8) is mounted in the forearm area and carries a set of encoders (9). The rotation of the encoders in the set (9) is rigidly linked to the finger movements via cables routed from the phalanges through the branching mechanism. Each encoder in the set (9) is configured to measure: the angular displacement occurring during flexion or extension of a specific phalanx; and the direction and magnitude of movement transmitted through the cable system from the finger to the corresponding encoder in the set (9).

Thus, the fastener (8) with the set of encoders (9) functions as a central measurement unit that converts the mechanical motion of the fingers into digital data for analysis.

The encoders used in the design of the claimed mechanical motion-capture glove may be of various types, depending on the required accuracy, operating conditions, and measurement characteristics. In particular, the device may utilize absolute, incremental, or other types of encoders.

The components of the claimed mechanical glove are structurally connected to each other, for example, by means of threaded connections (screwing), which ensures reliable fixation, the possibility of disassembly, and subsequent adjustment of individual units and components.

The following describes a second embodiment of the mechanical motion-capture glove — with force feedback.

The second embodiment of the claimed mechanical motion-capture glove for robotics is designed to be worn on the human wrist and is configured to measure and transmit kinematic parameters of finger flexion/extension and abduction/adduction movements, as well as to collect tactile information from the fingertips and detect wrist flexion/extension. The glove comprises the following functional components:

The first fastener (1) connects the distal and intermediate phalanges of the finger, forming a rigid or semi-rigid structure. A tactile sensor (2) is installed in the area of the distal phalanx of the fastener (1), designed to measure the pressure force applied by the fingertip during contact with objects. On the dorsal side of the phalanx of the first fastener (1), a cable attachment is provided, enabling the transmission and measurement of flexion and extension of the distal and intermediate phalanges.

The second fastener (3) is mounted on the proximal phalanx. On the dorsal side of the proximal phalanx of the second fastener (3), a cable attachment is provided, through which the flexion and extension of the proximal phalanx of the finger are transmitted and measured. In

addition, the second fastener (3) includes a rigid coupling designed to transmit mechanical motion to the first encoder (4). The first encoder (4) is configured to measure the rotation angle of the proximal phalanx relative to the base of the hand, thereby providing feedback during grasping and manipulation.

The first encoder (4) is located on the dorsal surface of the hand and is intended to measure finger abduction and adduction (i.e., lateral spreading and convergence relative to the hand axis). The measurement is performed via a rigid mechanical connection between the first encoder (4) and the second fastener (3) mounted on the proximal phalanx of the finger. This structural design enables the transmission of angular displacement during lateral finger movements, allowing for precise registration of the finger's position in the plane perpendicular to flexion and extension.

An identical structure, as described above, is provided for the remaining fingers of the hand. This enables full measurement of all primary degrees of freedom of each finger, including: flexion and extension of the distal and intermediate phalanges; flexion and extension of the proximal phalanx; abduction and adduction of the finger (movement in the lateral plane relative to the axis of the hand); and measurement of the pressure applied to the fingertip using integrated tactile sensors.

An additional second encoder (5) is provided for the thumb and is mounted on the dorsal side of the hand. The second encoder (5) is configured to measure the rotation of the thumb relative to the base of the hand, taking into account its unique anatomical mobility, including movement in the saddle joint—such as rotation, opposition, and abduction. The installation of the second encoder (5) provides an additional degree of freedom in the measurement system and allows for more accurate reproduction of thumb movements during object grasping and manipulation.

A branching mechanism (6) is located at the transition between the hand and the wrist. It serves to distribute the cables running from the finger phalanges to the actuation and measurement unit positioned on the forearm. The branching mechanism (6) ensures the routing of cables from each finger in such a way as to prevent cable crossing and excessive tension during wrist movement, while also reducing friction and wear through the use of guiding channels and rotary elements.

A third encoder (7) is mounted on the wrist and is configured to measure flexion and extension of the hand at the wrist joint. The third encoder (7) records the angular position of the hand relative to the forearm, allowing the influence of wrist position on the overall configuration

of the fingers and the entire hand to be taken into account. The presence of this sensor enables correction of finger flexion calculations based on the wrist flexion angle.

A third fastener (8) is mounted in the forearm area and carries a set of encoders (9). The rotation of the encoders in the set (9) is rigidly linked to the finger movements via cables routed from the phalanges through the branching mechanism. Each encoder in the set (9) is configured to measure: the angular displacement occurring during flexion or extension of a specific phalanx of the finger; and the direction and magnitude of movement transmitted through the cable system from the finger to the corresponding encoder in the set (9).

Thus, the third fastener (8) with the set of encoders (9) functions as a central measurement unit that converts the mechanical motion of the fingers into digital data for analysis.

To provide force (kinesthetic) feedback functionality, motors (10) are additionally installed on the encoders from the set of encoders (9) mounted on the third fastener (8) located on the forearm.

Each motor (10) is rigidly coupled to the shaft of the corresponding encoder and, consequently, to the cable transmitting motion to the phalanx of the finger. This configuration allows for the generation of adjustable resistance to finger movement, simulating interaction with virtual or physical objects (e.g., the sensation of hardness, elasticity, or range-of-motion constraints), and enables forced stopping or slowing of finger motion when a virtual object applies resistance.

The encoders used in the design of the claimed mechanical motion-capture glove may be of various types, depending on the required accuracy, operating conditions, and specific measurement characteristics. In particular, the device may employ absolute, incremental, and other types of encoders.

The components of the claimed mechanical glove are structurally connected to one another, for example, by means of threaded connections (screwing), which ensures reliable fixation, the possibility of disassembly, and subsequent adjustment of individual units and components.

Examples of implementation of the claimed technical solution are provided below.

Example 1. Use of the claimed mechanical motion-capture glove for robotics without force feedback (according to the first embodiment).

The claimed mechanical motion-capture glove is worn on the operator's hand and forearm, providing registration of finger and wrist movements. When the intermediate phalanx of one of the fingers is flexed, the cable attached to the dorsal side of the phalanx transmits the resulting mechanical displacement through the branching mechanism (6) to the corresponding encoder in the set of encoders (9) located on the third fastener (8) mounted on the forearm.

Since the cable is connected to the first fastener (1), which joins the distal and intermediate phalanges, movement of one element causes coordinated motion of the entire segment. As a result, the corresponding encoder in the set of encoders (9) converts the rotation into a digital signal corresponding to the flexion/extension angles of both the intermediate and distal phalanges.

At the moment when pressure is applied to the tip of one of the fingers, the tactile sensor (2) located on the distal phalanx registers the magnitude of the applied force. The sensor detects contact with an object and converts the pressure into an electrical signal corresponding to the level of force acting on the finger surface.

When the proximal phalanx of one of the fingers is flexed, the cable attached to the dorsal side of the phalanx transmits the resulting mechanical displacement through the branching mechanism (6) to the corresponding encoder from the set of encoders (9) mounted on the third fastener (8) on the forearm.

Since the cable is rigidly connected to the second fastener (3) mounted on the proximal phalanx, movement of the phalanx causes rotation of the corresponding encoder. This encoder converts the mechanical displacement into a digital signal corresponding to the flexion or extension angle of the proximal phalanx.

When one of the fingers is abducted (i.e., laterally moved away from the axis of the hand), the first encoder (4) mounted on the dorsal side of the hand registers the angular displacement of the corresponding finger.

Due to the rigid connection between the first encoder (4) and the fastener (3) located on the proximal phalanx of the finger, any lateral movement causes the encoder to rotate, converting this motion into a digital signal corresponding to the angle of finger abduction or adduction.

When the thumb rotates relative to the base of the hand, the second encoder (5) mounted on the dorsal side of the hand registers the angular displacement of the joint. The rotation of the second encoder (5) is converted into a digital signal corresponding to the thumb rotation angle.

During wrist movement (flexion or extension at the wrist joint), the encoder (7) rotates. This rotation is converted into a digital signal corresponding to the current wrist flexion angle.

The data obtained from the third encoder (7) is compared with the data from the encoders in the set (9) mounted on the third fastener (8) on the forearm, which register the movements of individual finger phalanges. This comparison makes it possible to correct kinematic errors that occur due to changes in wrist orientation when the fingers flex during simultaneous wrist movement, thereby ensuring accurate reconstruction of the entire hand position in space.

Thus, each finger movement is recorded independently, thanks to the separate measurement of flexion/extension and abduction/adduction angles of each phalanx using individual cables and corresponding encoders from the set (9) mounted on the third fastener (8) on the forearm.

Each finger is equipped with its own system of fasteners, cables, and sensors, which provides: independent registration of the kinematics of each joint (distal, intermediate, and proximal phalanges); accounting for lateral movements (abduction and adduction) using additional encoders; and acquisition of complete and continuous information on the movement of each finger, thereby eliminating mutual interference or interpretation errors during simultaneous movements of multiple fingers.

Example 2. Real-time robot control using the mechanical motion-capture glove with force feedback according to the second embodiment.

In the second example, the claimed mechanical motion-capture glove is used for direct real-time control of a robotic manipulator.

The operator puts the glove on their hand and connects it to the robot control system. When performing finger and wrist movements—such as grasping, releasing, flexion, abduction, or thumb rotation—the mechanical displacements are transmitted via the cable system to the encoders in the set of encoders (9) located on the fastener (8) on the forearm. The encoders convert the bending angles of the phalanges and joint positions into digital signals.

The third encoder (7) on the wrist registers wrist flexion; the second encoder (5) on the thumb registers its rotation and opposition; the tactile sensors (2) on the fingertips detect pressure and contact force.

All data from the encoders and tactile sensors are processed by the control controller and transmitted as control commands to the robot's actuators. If the robot is equipped with tactile sensors mounted on its fingertips, the feedback signals activate the motors (10) installed on the corresponding encoders, generating adjustable resistance to the operator's finger movements.

This provides force feedback functionality and enables the robot to accurately replicate the user's hand movements with a high degree of synchronization, sensitivity, and realistic interaction.

Example 3. Collection of kinematic data for training a robot control system using the mechanical motion-capture glove without force feedback according to the first embodiment.

In the third example, the claimed mechanical motion-capture glove is used as a means for capturing and recording the operator's hand kinematic data for subsequent training of the robot control system (e.g., using machine learning methods, dynamic modeling, or simulation-based training).

The operator performs a series of targeted hand movements, such as grasping, moving, manipulating various objects, as well as precise and smooth motions. During this process: The encoders mounted on the fastener on the forearm record the flexion and extension angles of the phalanges of all fingers; The encoders on the wrist and thumb record movements in the respective joints; The tactile sensors on the fingertips collect data on contact pressure; All data are synchronously recorded and timestamped.

The obtained data form a training dataset, including:

- angular positions of the phalanges;
- spatial position of the hand;
- pressure force on the fingertips;
- trajectories of hand movements.

This dataset is used for:

- training neural network or hybrid robot control models (e.g., for a "humanoid" manipulator);
- constructing kinematic and dynamic models of hand movement;
- synthesizing realistic animations or virtual avatars in VR/AR environments.

Thus, the glove serves as a tool for collecting reference movements necessary for programming, training, or calibrating a robotic system based on real human actions.

Example 4. Interactive control of a virtual avatar or animated character using the mechanical motion-capture glove with force feedback according to the second embodiment.

In this example, the claimed mechanical motion-capture glove is used for interactive control of a virtual avatar or animated character in virtual or augmented reality environments, as well as in 3D animation systems.

The user wears the glove connected to a computer or VR system. When performing hand movements—such as gestures, finger expressions, touches, and manipulations—the data from the glove is transmitted to software, where:

Encoders record precise flexion/extension and abduction/adduction angles of the phalanges of all fingers; Encoders on the wrist and thumb register the overall position of the hand and complex movements such as rotation, tilt, and opposition; Motors installed on the actuation units are used to generate force feedback, allowing the operator to feel resistance and interaction with objects in virtual or physical environments.

The processed data is transmitted in real time to the virtual avatar's engine system, enabling:

- the avatar to replicate the user's movements with high accuracy, including complex finger expressions;
- the use of gestures as a means of interface control;
- intuitive interactions with virtual objects, including tactile force feedback.

Thus, based on the foregoing, it can be concluded that the **applicant has solved the identified technical problem and achieved the claimed technical result:** the development of a Mechanical Motion-Capture Glove for Robotics (Variants), providing implementation of:

- detection of wrist flexion, which is necessary for accurate calculation of finger flexion angles taking into account changes in palm orientation in space;
- collection of tactile information from the fingertips, enabling measurement of pressure force, touch, and interaction with objects;
- full capture of finger movements, including abduction and adduction, with the ability to reconstruct individual trajectories of each finger.

Claims

1. A Mechanical Motion-Capture Glove for Robotics, configured to be worn on a human wrist, **comprising:**

For each finger, a first fastener connecting the distal and intermediate phalanges of the finger, wherein a tactile sensor is installed in the region of the distal phalanx of the first fastener, the sensor being configured to measure the pressure force applied by the fingertip during contact with objects; and a cable attachment provided on the dorsal side of the phalanx of the first fastener, configured to transmit and measure flexion and extension of the distal and intermediate phalanges;

For each finger, a second fastener mounted on the proximal phalanx, wherein a cable attachment is provided on the dorsal side of the proximal phalanx of the second fastener, configured to transmit and measure flexion and extension of the proximal phalanx; wherein the second fastener includes a rigid coupling configured to transmit mechanical motion to a first encoder, the encoder being configured to measure the rotation angle of the proximal phalanx relative to the base of the hand, thereby providing feedback during grasping and manipulation;

For each finger, a first encoder located on the dorsal surface of the hand and configured to measure finger abduction and adduction, wherein the measurement is carried out via a rigid mechanical connection between the first encoder and the second fastener mounted on the proximal phalanx of the finger, enabling transmission of angular displacement during lateral finger movements and precise registration of the finger's position in the plane perpendicular to flexion and extension;

An additional second encoder mounted on the dorsal side of the hand for the thumb, the second encoder configured to measure rotation of the thumb relative to the base of the hand, taking into account its anatomical mobility including movement in the saddle joint such as rotation, opposition, and abduction, thereby providing an additional degree of freedom in the measurement system and enabling accurate reproduction of thumb movements during grasping and manipulation of objects;

A branching mechanism located at the transition between the hand and the wrist, configured to distribute cables running from the finger phalanges to the actuation and measurement unit positioned on the forearm, wherein the branching mechanism ensures routing of cables from each finger to prevent cable crossing and excessive tension during wrist movement, and reduces friction and wear of the cables through guiding channels and rotary elements;

A third encoder mounted on the wrist, configured to measure flexion and extension of the hand at the wrist joint, to record the angular position of the hand relative to the forearm, to account for the influence of wrist position on the overall position of the fingers and the entire hand, and to provide correction of finger flexion calculations based on the wrist flexion angle;

A third fastener mounted on the forearm, carrying a set of encoders, the rotation of which is mechanically linked to finger movements via cables routed from the phalanges through the branching mechanism, wherein each encoder in the set is configured to measure: angular displacement occurring during flexion or extension of a specific phalanx of a finger; the direction and magnitude of movement transmitted through the cable system from the finger to the corresponding encoder in the set.

2. A Mechanical Motion-Capture Glove for Robotics, configured to be worn on a human wrist, **comprising**, for each finger:

For each finger, a first fastener connecting the distal and intermediate phalanges of the finger, wherein a tactile sensor is installed in the region of the distal phalanx of the first fastener, the sensor configured to measure pressure force applied by the fingertip during contact with objects; and a cable attachment provided on the dorsal side of the phalanx of the first fastener, configured to transmit and measure flexion and extension of the distal and intermediate phalanges;

For each finger, a second fastener mounted on the proximal phalanx, wherein a cable attachment is provided on the dorsal side of the proximal phalanx of the second fastener, configured to transmit and measure flexion and extension of the proximal phalanx; wherein the second fastener includes a rigid coupling configured to transmit mechanical motion to a first encoder, the encoder configured to measure a rotation angle of the proximal phalanx relative to the base of the hand, thereby providing feedback during grasping and manipulation;

For each finger, a first encoder located on the dorsal surface of the hand and configured to measure finger abduction and adduction, wherein the measurement is performed via a rigid mechanical connection between the first encoder and the second fastener mounted on the proximal phalanx of the finger, enabling transmission of angular displacement during lateral finger movements and precise registration of the finger's position in the plane perpendicular to flexion and extension;

An additional second encoder mounted on the dorsal side of the hand for the thumb, the second encoder configured to measure rotation of the thumb relative to the base of the hand, taking into account its anatomical mobility including movement in the saddle joint, such as rotation, opposition, and abduction, thereby providing an additional degree of freedom in the

measurement system and enabling accurate reproduction of thumb movements during grasping and manipulation of objects;

A branching mechanism located at the transition between the hand and the wrist, configured to distribute cables running from the finger phalanges to the actuation and measurement unit positioned on the forearm, the branching mechanism configured to route cables from each finger to prevent cable crossing and excessive tension during wrist movement, and to reduce friction and wear of the cables via guiding channels and rotary elements;

A third encoder mounted on the wrist, configured to measure flexion and extension of the hand at the wrist joint, to record the angular position of the hand relative to the forearm, to account for the influence of wrist position on the overall position of the fingers and the entire hand, and to provide correction of finger flexion calculations based on the wrist flexion angle;

A third fastener mounted on the forearm, carrying a set of encoders mechanically linked to finger movements via cables routed from the phalanges through the branching mechanism, wherein each encoder in the set is configured to measure: angular displacement occurring during flexion or extension of a specific phalanx of a finger; and the direction and magnitude of movement transmitted through the cable system from the finger to the corresponding encoder in the set;

Wherein, to provide force feedback functionality, motors are additionally installed on the encoders from the set of encoders mounted on the third fastener on the forearm, each motor being rigidly coupled to the shaft of the corresponding encoder and thus to the cable transmitting motion to the finger phalanx, enabling adjustable resistance to finger movement to simulate interaction with virtual or physical objects, and allowing forced stopping or slowing of finger motion.

ABSTRACT

Mechanical Motion-Capture Glove for Robotics (Variants)

The present inventions relate to the fields of robotics and mechatronics, specifically to wearable exoskeleton devices. The objective of the present invention is to create a mechanical glove kinematically identical to the anatomical structure of the human hand, designed for simultaneous real-time control of a robotic hand and for capturing kinematic parameters of the user's finger movements for subsequent training and programming of robotic systems.

The essence of the claimed technical solution is a Mechanical Motion-Capture Glove for Robotics, designed to be worn on a human wrist, **comprising:** For each finger, a first fastener connecting the distal and intermediate phalanges of the finger, wherein a tactile sensor is installed in the region of the distal phalanx of the first fastener, the sensor configured to measure the pressure force applied by the fingertip during contact with objects; a cable attachment is provided on the dorsal side of the phalanx of the first fastener, configured to transmit and measure flexion and extension of the distal and intermediate phalanges; For each finger, a second fastener mounted on the proximal phalanx, wherein a cable attachment is provided on the dorsal side of the proximal phalanx of the second fastener, configured to transmit and measure flexion and extension of the proximal phalanx; the second fastener includes a rigid coupling configured to transmit mechanical motion to a first encoder, the encoder configured to measure the rotation angle of the proximal phalanx relative to the base of the hand, thereby providing feedback during grasping and manipulation; For each finger, a first encoder located on the dorsal surface of the hand and configured to measure finger abduction and adduction, wherein the measurement is performed via a rigid mechanical connection between the first encoder and the second fastener mounted on the proximal phalanx, enabling transmission of angular displacement during lateral finger movements and precise registration of the finger's position in the plane perpendicular to flexion and extension; For the thumb, an additional second encoder mounted on the dorsal side of the hand, configured to measure rotation of the thumb relative to the base of the hand, taking into account its anatomical mobility, including movements in the saddle joint such as rotation, opposition, and abduction, thereby providing an additional degree of freedom in the measurement system and enabling accurate reproduction of thumb movements during grasping and manipulation of objects; A branching mechanism located at the transition between the hand and the wrist, configured to distribute cables running from the finger phalanges to the actuation and measurement unit positioned on the forearm, ensuring cable routing from each finger to prevent cable crossing and excessive tension during wrist movement, and reducing friction and wear of the cables via guiding channels and rotary elements; A third encoder mounted on the

wrist, configured to measure flexion and extension of the hand at the wrist joint, record the angular position of the hand relative to the forearm, account for the influence of wrist position on the overall position of the fingers and entire hand, and provide correction of finger flexion calculations based on the wrist flexion angle; A third fastener mounted on the forearm, carrying a set of encoders mechanically linked to finger movements via cables routed from the phalanges through the branching mechanism, wherein each encoder in the set is configured to measure angular displacement occurring during flexion or extension of a specific phalanx of a finger, as well as the direction and magnitude of movement transmitted through the cable system from the finger to the corresponding encoder.

A Mechanical Motion-Capture Glove for Robotics, designed to be worn on a human wrist, **comprising:** For each finger, a first fastener connecting the distal and intermediate phalanges of the finger, wherein a tactile sensor is installed in the region of the distal phalanx of the first fastener, the sensor configured to measure the pressure force applied by the fingertip during contact with objects; a cable attachment is provided on the dorsal side of the phalanx of the first fastener, configured to transmit and measure flexion and extension of the distal and intermediate phalanges; For each finger, a second fastener mounted on the proximal phalanx, wherein a cable attachment is provided on the dorsal side of the proximal phalanx of the second fastener, configured to transmit and measure flexion and extension of the proximal phalanx; the second fastener includes a rigid coupling configured to transmit mechanical motion to a first encoder, the encoder configured to measure the rotation angle of the proximal phalanx relative to the base of the hand, thereby providing feedback during grasping and manipulation; For each finger, a first encoder located on the dorsal surface of the hand and configured to measure finger abduction and adduction, wherein the measurement is performed via a rigid mechanical connection between the first encoder and the second fastener mounted on the proximal phalanx, enabling transmission of angular displacement during lateral finger movements and precise registration of the finger's position in the plane perpendicular to flexion and extension; For the thumb, an additional second encoder mounted on the dorsal side of the hand, configured to measure rotation of the thumb relative to the base of the hand, taking into account its anatomical mobility including movements in the saddle joint such as rotation, opposition, and abduction, thereby providing an additional degree of freedom in the measurement system and enabling accurate reproduction of thumb movements during grasping and manipulation of objects; A branching mechanism located at the transition between the hand and the wrist, configured to distribute cables running from the finger phalanges to the actuation and measurement unit positioned on the forearm, ensuring cable routing from each finger to prevent cable crossing and

excessive tension during wrist movement, and reducing friction and wear of the cables via guiding channels and rotary elements; A third encoder mounted on the wrist, configured to measure flexion and extension of the hand at the wrist joint, to record the angular position of the hand relative to the forearm, to account for the influence of wrist position on the overall position of the fingers and entire hand, and to provide correction of finger flexion calculations based on the wrist flexion angle; A third fastener mounted on the forearm, carrying a set of encoders mechanically linked to finger movements via cables routed from the phalanges through the branching mechanism, wherein each encoder in the set is configured to measure angular displacement occurring during flexion or extension of a specific phalanx of a finger, as well as the direction and magnitude of movement transmitted through the cable system from the finger to the corresponding encoder; To provide force feedback functionality, motors are additionally installed on the encoders from the set of encoders mounted on the third fastener on the forearm, each motor being rigidly coupled to the shaft of the corresponding encoder and thus to the cable transmitting motion to the finger phalanx, enabling adjustable resistance to finger movement to simulate interaction with virtual or physical objects, and allowing forced stopping or slowing of finger motion.

2 independent claims, 3 figures.

Reference: F.R. Khabibullin

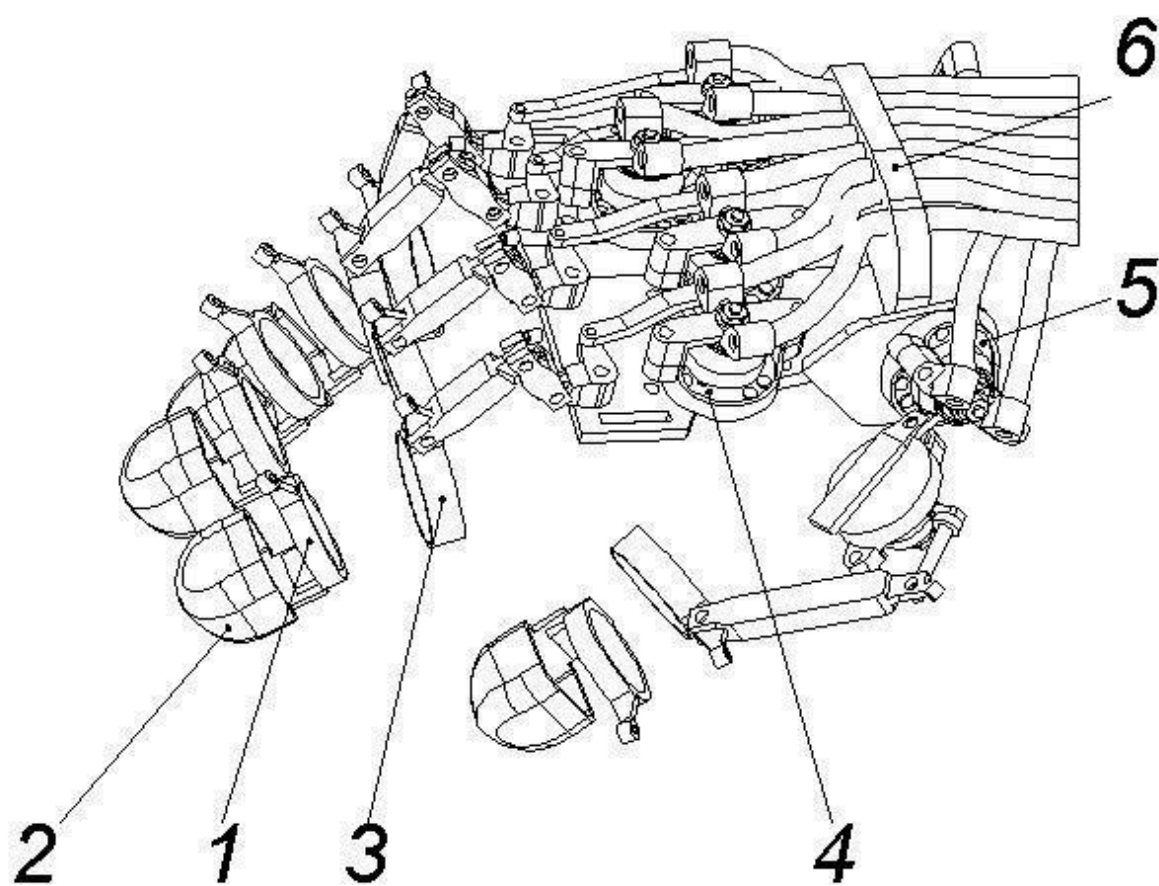


Fig.1

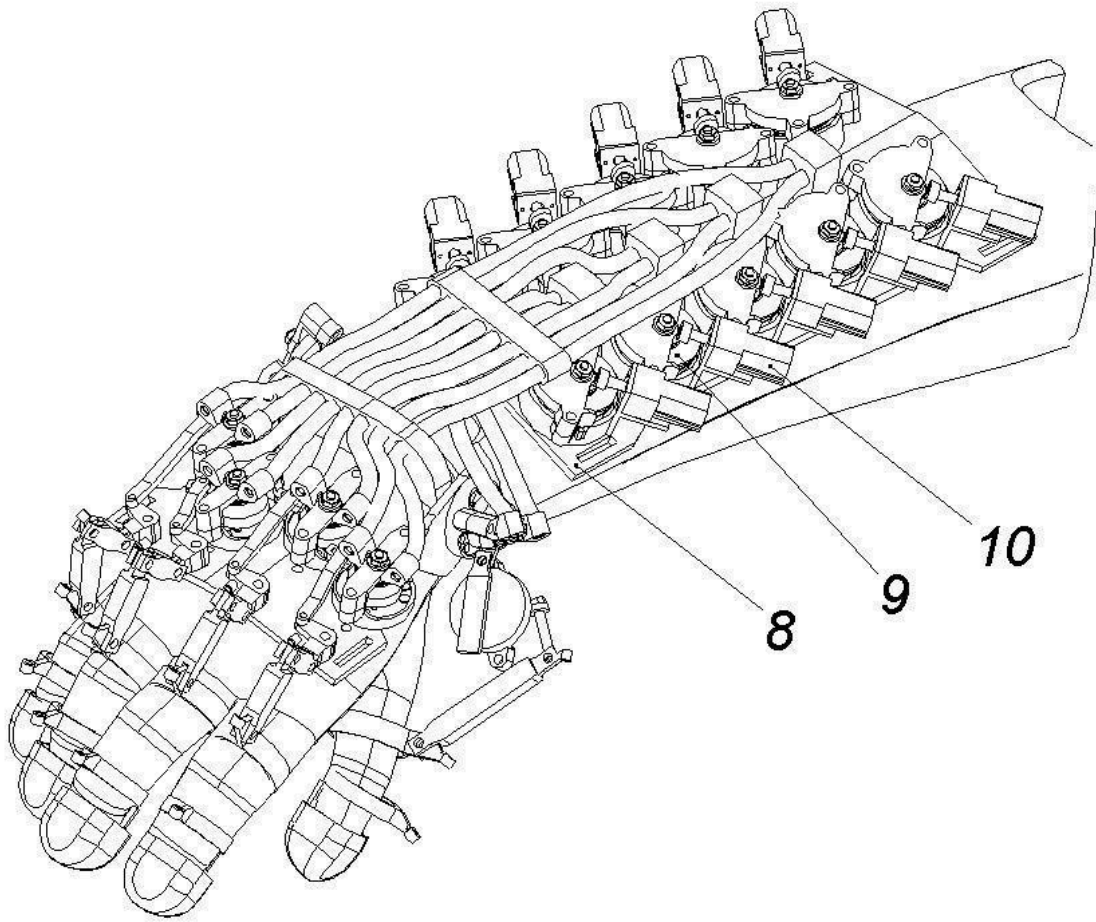


Fig.2

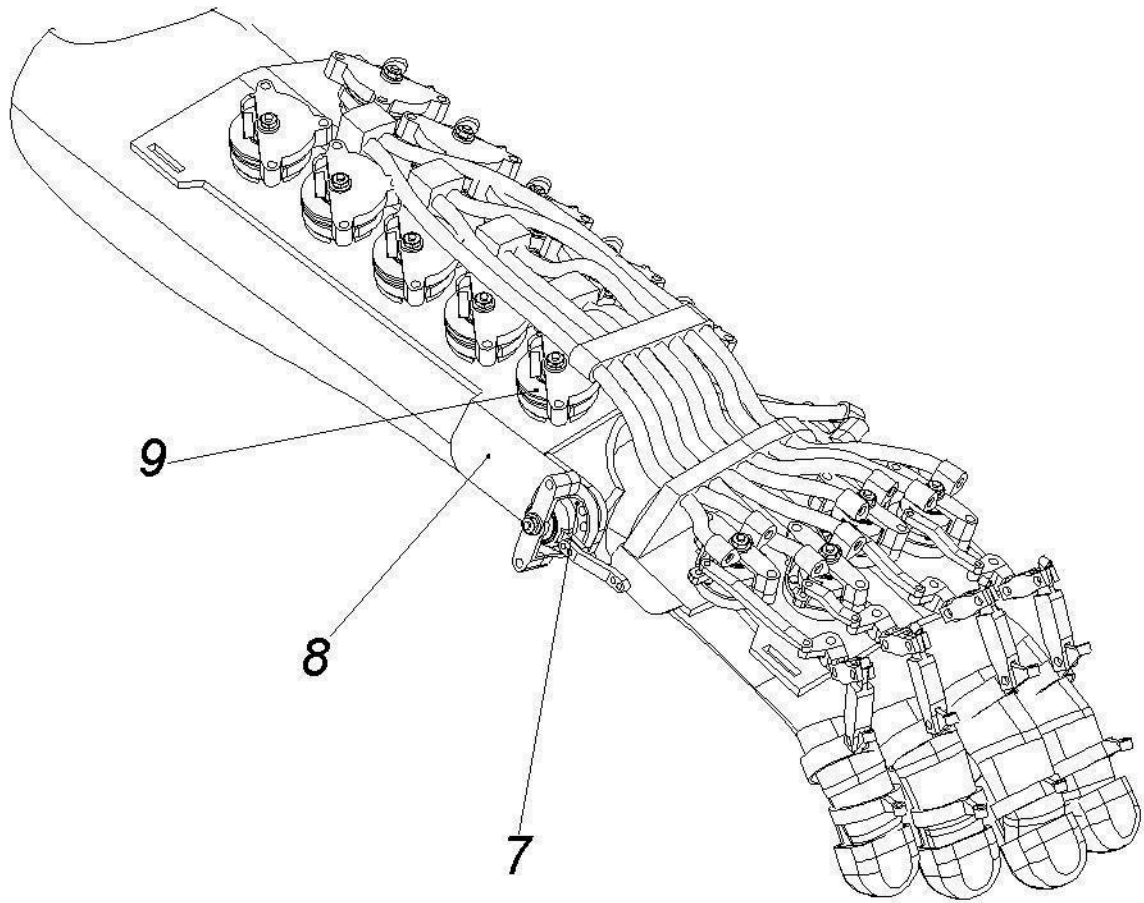


Fig.3